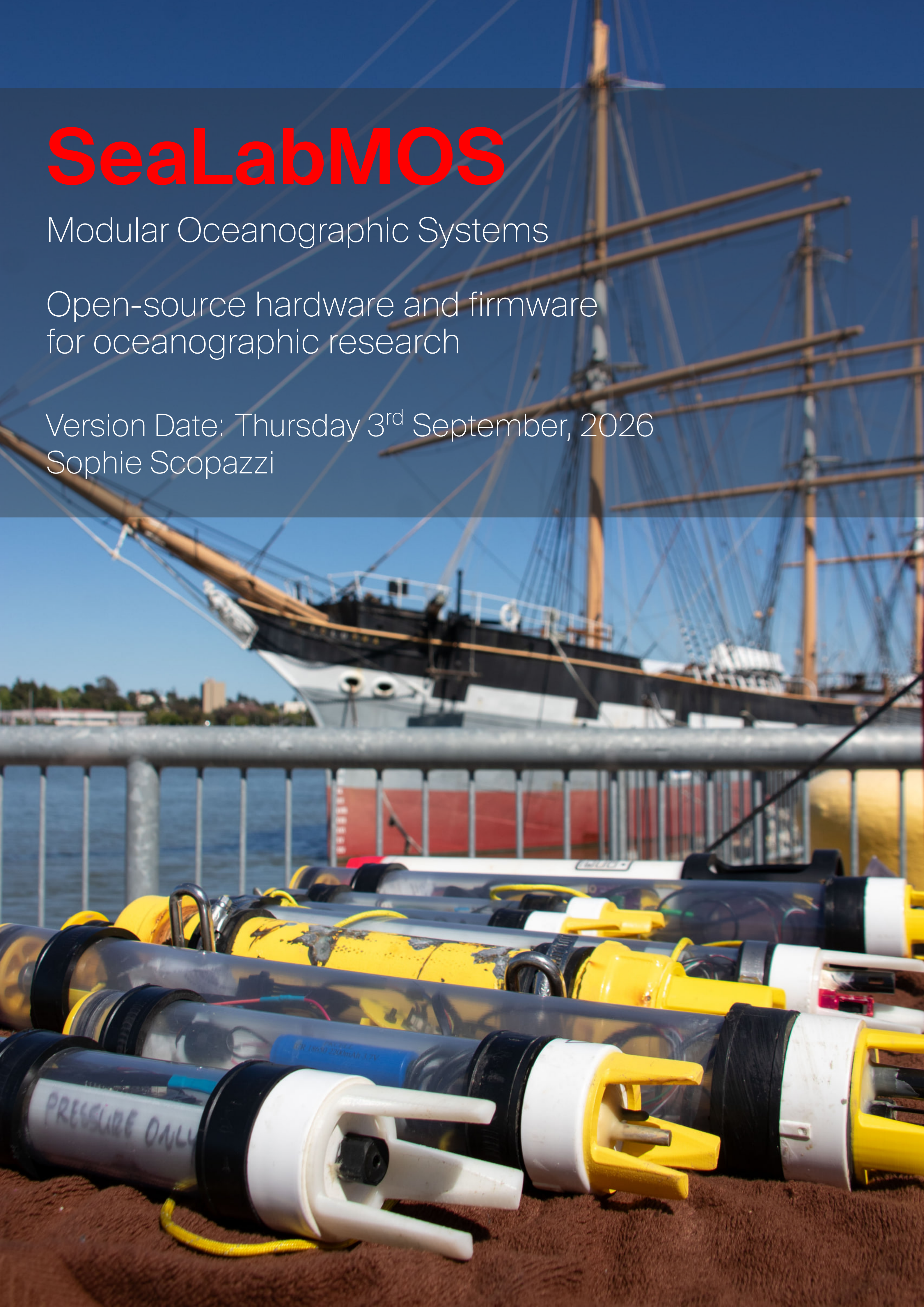
SeaLabMOS

Modular Oceanographic Systems

Open-source hardware and firmware
for oceanographic research

Version Date: Thursday 3rd September, 2026

Sophie Scopazzi



Contents

Preface

This manual remains a work in progress. Please look at version date below.
If you have experience with oceanographic sampling systems/methods, I recommend skipping to chapter 2.
If you only want the wiring instructions, see chapter 3
If you have any questions, please reach out to me!

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0.1. SeaLabMOS System Configurations

The system has been designed with two operational methods in mind: profiling and static. As the SeaLabMOS family of systems are designed to be modular, each depoyment method has different sensors that better suit each type. When looking at the following table think about the science questions you are attempting to answer.

1. What do you want to sample (temperature, salinity, location)
2. How deep do you you want to sample?
3. Do you want to see live data at the surface or at depth?
4. How long will the system be deployed for (the desired endurance)?

Table 1: SeaLabMOS Deployment Types and Platforms

Profile			**Static**			**Live View**
			Bottom Pressure Recorder	**Temperature Logger**	**Surface**	**Wired Profile**
1.5 in PVC	2 in PVC	2 in Steel	1.5 in PVC	1.5 in PVC	Nalgene	Pelican case
RP2040	RP2040	RP2040	RP2040	RP2040	RP2040	Raspberry Pi
–	–	–	–	–	GNSS (option)	GNSS
RTC	RTC	RTC	RTC	RTC	RTC/GNSS	GNSS Time
OLED	OLED	OLED	–	–	–	Screen
Switch	Switch	Switch	Switch	Switch	Switch	12 V to 5 V
2.2 Ah	4.4 Ah	4.4 Ah	4.4 Ah	2.2 Ah	4.4, 6.6, 10 Ah	–
–	–	–	–	–	Solar (option)	–
Salt	Salt	Salt	–	–	Salt	Salt
BR Temp	BR Temp	BR Temp	–	–	–	BR Temp
BR Bar30	BR Bar30	BR Bar100	BR Bar02	–	–	BR Bar30
–	Thermistor	Thermistor	–	Thermistor	Thermistor	Thermistor
–	DS18B20	DS18B20	–	DS18B20	DS18B20	DS18B20
–	–	–	–	–	Orange LED	–
140m	140m	350m	140m	140m	2m	140 m–350m

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Ocean Sampling Background

Chapter Concepts:

1. What are different types of oceanographic sensor platforms and what are they used for?
2. How are research platforms typically deployed?

1.1. What is a CTD?

CTD (for **C**onductivity, **T**emperature, and **D**epth) refers to the three primary parameters measured by the instrument. Salinity is not measured directly but sampled with a conductivity sensor then calculated using the in-situ tvertical emperature and pressure. Data from a CTD can be used to plot salinity and temperature profiles as a function of depth. An example of what this can look like is Fig. 1.1.

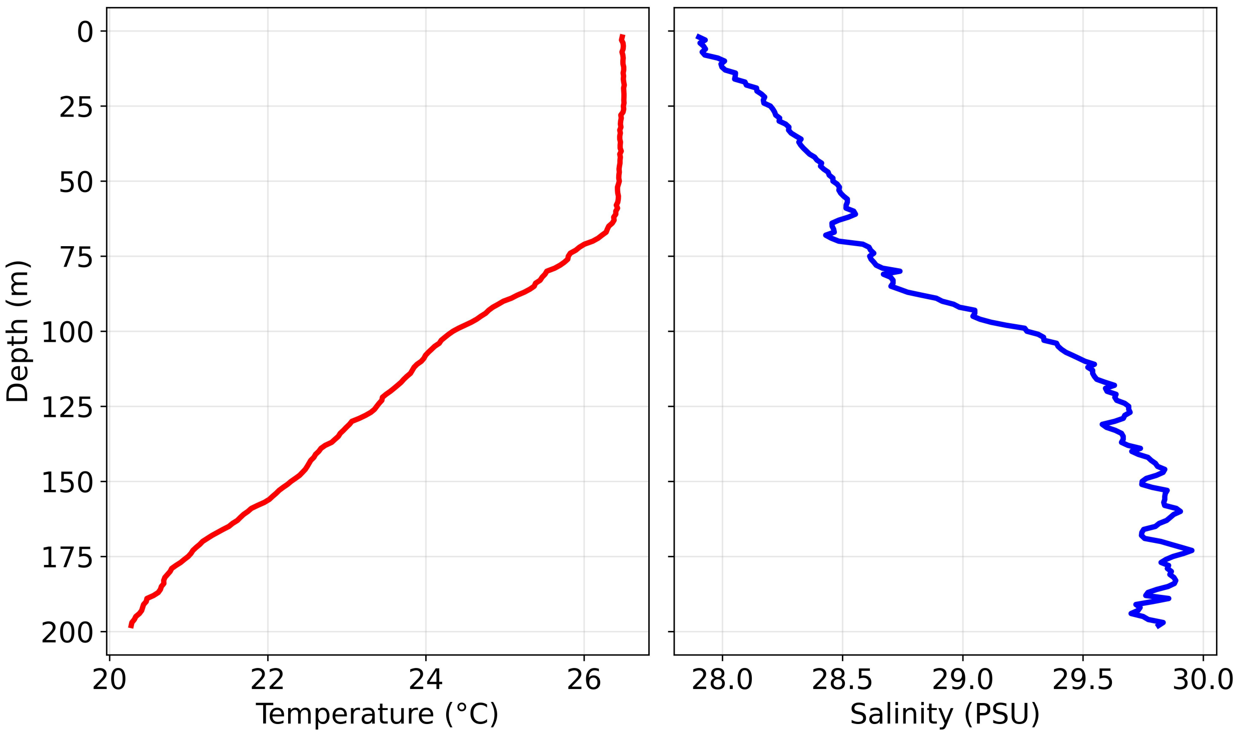

Figure 1.1: 200 meter cast of the two-inch steel SeaLabMOS CTD

Another common method to display CTD data is called a TS diagram (short for temperature-salinity) because the plot compares the two variables, typically using color to indicate depth. Fig. 1.2 shows that the surface water was warmer and less salty, with temperature decreasing and salinity increasing with depth.

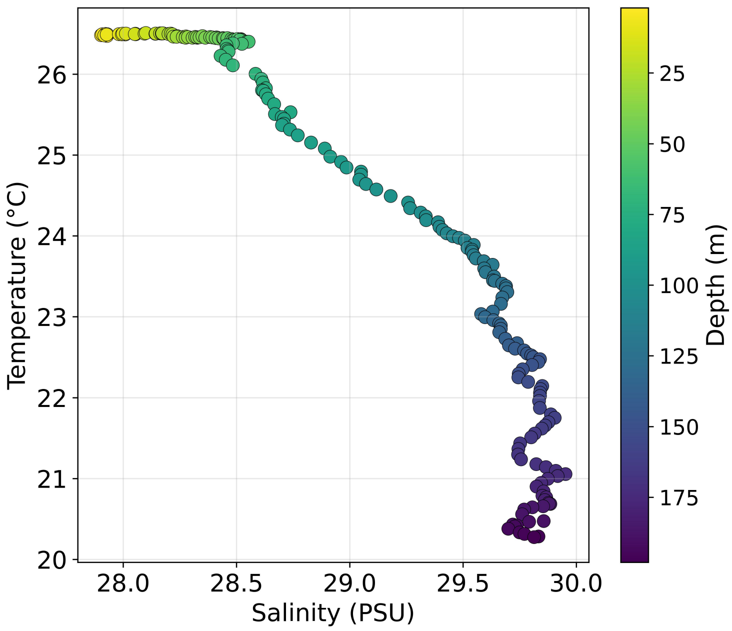

Figure 1.2: Temperature–Salinity (TS) diagram of the same data as in Fig. 1.1

1.2. CTD Applications

CTD data can be used to answer questions such as whether the water column is stable and resistant to vertical mixing or unstable and prone to mixing, and to determine the depth of the thermocline or halocline. This information also assists with further study of the micro-organisms in the epipelagic zone, the upper photic layer of the ocean where sunlight penetrates.

Another important application of CTD data is calculating the speed of sound through the water column, as water temperature and salinity effect the density of seawater, which directly effects sound propagation through the medium. The variation in water density and temperature bend or reflect sound waves, making this also important information for areas of naval warfare such as detection range and target location accuracy. Beyond these applications, understanding sound propagation in the water column is critical for underwater communication systems, environmental monitoring, and scientific surveys, where accurate acoustic measurements are necessary for multi-beam sonar mapping of the seafloor.

1.3. How CTDs are deployed

Generally speaking, a oceanographic research platform can be deployed dynamically or statically. A system can be...

1. Lowered from the surface to vertically profile a column of water, typically called a cast. It could also be affixed to a system that traverses the water column up and down (a winch, wire walker, ocean glider, Argo float, etc.).
2. Affixed to a surface buoy, pier, anchored to the sea floor, or other similarly static location.
3. On a drifting surface float.

Each deployment type has different design considerations, sensor selection, and resulting cost trade-offs. For example, a stationary moored system can be powered via a cable with power-hungry sensors with a fast swample rate, while a mobile profiling float prioritizes compactness, low weight, and energy-efficient data logging. When it comes to sampling the ocean, there are many possible system configurations, ranging from simple single-parameter sensors to fully integrated, multi-sensor platforms capable of measuring temperature, salinity, dissolved oxygen, turbidity, and more. Each configuration presents trade-offs in terms of data resolution, deployment duration, maintenance requirements, and overall cost, meaning the choice of system must align closely with the scientific goals, operational constraints, and environmental conditions of the deployment.

1.4. Temperature Loggers

Uses:

- Monitor marine heatwaves and long-term climate trends.
- Characterize thermal stratification and mixing in the water column.

- Support habitat studies for corals, kelp, and fish.
- Provide baseline environmental data for impact assessments.
- Validate ocean models and satellite sea-surface temperature products.

1.5. Bottom Pressure Recorder

Uses:

- Measure water level changes for tidal and tsunami monitoring.
- Track long-term sea-level rise in coastal and deep-ocean sites.
- Observe internal waves, storm surges, and barotropic currents.
- Support calibration of numerical ocean circulation models.
- Provide reference pressure data for moorings and CTD deployments.

1.6. Surface Float

Uses:

- Sample the surface ocean for temperature, salinity, etc., while also potentially carrying meteorological sensors such as wind, temperature, air pressure, and humidity.
- Keep moorings vertical and help maintain sensor depth.
- Provide surface visibility and recovery aids for systems that are deployed underwater.
- Support GPS, Iridium, or LoRa telemetry packages for real-time data.
- Act as platforms for surface drifters or wave-measurement systems.

2

Base Hardware Components

Chapter Concepts:

1. Sensor selection considerations.

2.1. Microcontroller Selection and Base System

The SeaLabMOS main board is the Adafruit Feather RP2040 Adalogger - 8MB Flash with microSD Card. It has an onboard microSD card slot, battery charging capability, and STEMMA QT port for solderless sensor integration. The RP2040 microcontroller has the capability to be overclocked for additional performance and underclocked for additional power savings when endurance is paramount.

Many types/brands of microcontroller could be used, so long as they have an I2C bus and UART capability. Alternatives that may work include:

– Arduino Uno R3 / Adafruit Metro 328. Only fits in larger enclosures and does not have a built in microSD card, but has many shield options. Could be good for bench testing.
– Adafruit QT Py RP2040 (Product ID: Product ID: 4900[1]) with Adafruit LiIon or LiPoly Charger BFF Add-On for QT Py (Product ID: 5397[2]). *Extremely* small with no built in microSD card.
– Teensy 4.1[3]. Small and powerful. Probably overkill for the current SeaLabMOS use-case unless adapting this work into something more computationally intensive.

The required base parts of the SeaLabMOS are listed in Table 2.1. The OLED display with buttons, GPS/GNSS, and beacon are mission specific and used when needed, resulting in a cost range:

– $45 no screen
– $60 with screen
– $90 with Mini GPS for location
– $111.5 with Ultimate GPS for location w/ external antenna

2.2. Firmware

The firmware is the same for all versions of the SeaLabMOS family of systems. The sensors are selected by boolean values in the main .ino file as defined in GitHub[4].

2.2.1. Default Record Format for Comma-Separated-Values

Due to the variable nature of SeaLabMOS, a default header format is used across all firmware. Each firmware type updates only the relevant attached sensor values while leaving the others untouched, and the nthe CSV is only populated by the updated values. In this manner, post-processing of data (or live data processing, if building a tethered system) is standardized with one data-ingestion header format.

[1] https://www.adafruit.com/product/4900
[2] https://www.adafruit.com/product/5397
[3] https://www.pjrc.com/store/teensy41.html
[4] https://github.com/sscopazzi/SeaLabMOS

Table 2.1: Base Datalogger Hardware

Name	Adafruit Part #	What it does	Accuracy	Cost
Adafruit Feather RP2040 Adalogger w/ microSD	5980	Datalogger	—	$15
SD/MicroSD Memory Card (512 MB)	5252	Data storage	—	$5
DS3231 Precision RTC FeatherWing	3028	Real time clock	±2 minutes per year	$14
CR1220 12mm 3 V Lithium Coin Cell Battery	380	Battery for RTC	—	$1
Lithium Ion Battery Pack 3.7 V 2.2 Ah (or 2.5–10 Ah)	1781 (2.2 Ah)	Battery	—	$10
Adafruit FeatherWing 128×64 OLED display	4650	Screen, Buttons	—	$15
Mini GPS (PA1010D)	4415	GPS/GNSS, Time	2 m, 10 Hz	$30
Ultimate GPS Breakout (PA1616D)	5440	GPS/GNSS, Time	**meters**, 10 Hz	$30
External Active GPS Antenna	960	For Ultimate	accuracy	$21.5
High-strength ’rare earth’ magnet	9	Hold antenna		$2.5
STEMMA QT / Qwiic JST SH 4-pin Cable	5385	Sensor Connection		$1–1.5
Stemma QT / Qwiic 5 Port Hub& 5625	Sensor Connection			$2.5

Note: Accuracy of RTC from manufacturer. GPS/GNSS module accuracy from testing.

2.2.2. Saving Data to the Onboard MicroSD Card

All data is indexed by time from the RTC and saved to the onboard microSD card as CSV, with newlines for each record. For dynamic systems, the firmware creates a new file at each power-on (e.g., 2025-07-10T14-22-17.csv), while the firmware for static systems for multi-day deployments creates one file per day (e.g., 2025-07-21.csv). The datetime format follows ISO 8601[5], with ”-” replacing ”:” to avoid forbidden filename characters.

Profile firmware at 200 MHz (5 Hz) produces 380–450 bytes/s (about 76–90 bytes/sample), filling a 256 MB card in 7–8 days. Static deployments sample far less frequently, so battery life, not file size, is the limiting factor. Using an 8 GB card removes storage concerns and smaller cards are feasible if sample duration is altered.

Saving to the onboard microSD restricts data access to post-deployment retrieval; live monitoring below the surface is not possible. Future wired version(s) of the SeaLabMOS are planned for real-time data monitoring of casts to depth.

2.3. Endurance and Power Monitoring

System endurance is highly variable, mostly due to the chosen battery. To assist in knowing what the battery voltage is upon system power-up, the following table is in the firmware to blink the Feather’s onboard NeoPixel. Section 3.1 contains the wiring information.

Table 2.2: Battery Voltage Ranges and Corresponding LED Colors

Voltage Range (V)	Color	Behavior	Description
$\geq$ 4.00	Green	Solid	Fully charged
3.80 – 3.99	Cyan	Solid	Mostly full
3.60 – 3.79	Yellow	Solid	Moderate
3.40 – 3.59	Purple	Solid	Low
3.35 – 3.39	Red	Solid	Critically low
$<$ 3.35	Red	Blinking	Dangerously low

2.3.1. Power Use of Each Component

This is difficult to measure without specialized equipment.

2.4. Real-Time Clock

For precise timekeeping, SeaLabMOS uses the DS3231 Precision RTC FeatherWing. The I2C-integrated Real Time Clock (RTC) fits right on top of the base board and has a Temperature Compensated Crystal

[5] **ISODates**.

Oscillator (TCXO) to help maintain accurate timing. The board also has a SQW/INT breakout pin, which is utilized to wake the microcontroller from a low-lower sleep mode.

2.5. OLED Display

If real-time data viewing, in-field pre-deployment button functionality and/or error message monitoring via text is desired, SeaLabMOS can integrate the Adafruit FeatherWing OLED - 128x64 OLED[6]. The OLED board also has an extra STEMMA QT connector.

2.6. Location (GPS/GNSS)

If location of the start of the CTD cast is desired, or if building a surface asset, the Adafruit Mini GPS PA1010D[7] can be integrated through the STEMMA QT port. If this board is integrated into the system and used for location and time, the RTC may not be required. Alternatively, the GNSS board could be utilized for location and time and to set the RTC time before entering ”sleep mode” (the sensor datasheet terms it Backup Mode) to conserve power.

2.7. Flashing Yellow Lighted Beacon

In the SeaLabMOS underwater configuration an LED beacon is not needed; however, it should be considered for surface assets that will be deployed during hours of darkness and reduced visibility in areas where mariners or others on the water may come across the platform.

Table 2.3: Components Used for Yellow LED Beacon

Component	Specification / Description	Link
Amber (590 nm) Rebel LED on a SinkPAD-II with Side Emitting Optic	74 lm @ 350 mA	luxeonstar.com[8]
350 mA BuckToot LED Driver – With Leads	Protects the LED	luxeonstar.com[9]
Adafruit 12 V Bias Voltage Boost Converter – TPS61040	Boosts 3.3 V from micrcontroller to 12 V for LED, has Enable pin for on/off	adafruit.com[10]

To maintain international standards for the safety of navigation, the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) maintains standards for navigational beacons. SeaLabMOS systems, being a scientific device, is within the category of special marks requiring a yellow light. A light must be present during all times of darkness and reduced visibility (fog, smoke, etc.).

2.7.1. Color

The color of the light (590 nm) is important. As stipulated within *Marine Signal Lights Colours E200-1* from December 2018 [**iala_levels_g1038_2016**], IALA strictly denotes the exact color of each light type. All “colours are specified with CIE 1931 standard colorimetric observer (2°-observer)” using Chromaticity Corner Coordinates (CCCs). The document indicates yellow using CCCs as stated in Table 2.4.

Table 2.4: IALA CCCs X and Y vs Nanometer Equivalent

	X	Y	Nanometer Equivalent
IALA Spec	0.5865	0.413	???.? (591.97)
Calculated	0.5752	0.424	590.0

The LED selected emits a wavelength of 590 nm, which falls within the yellow light specification required in Annex A, Color Regions.

2.7.2. Characteristic (Pattern)

As stipulated by IALA in 0110 Rhythmic Characters Of Lights On Marine Aids To Navigation[11], a special mark may only have certain characteristics. To simplify

[6] https://www.adafruit.com/product/4650
[7] https://www.adafruit.com/product/4415
[11] **iala__characters_r0110_2021**.

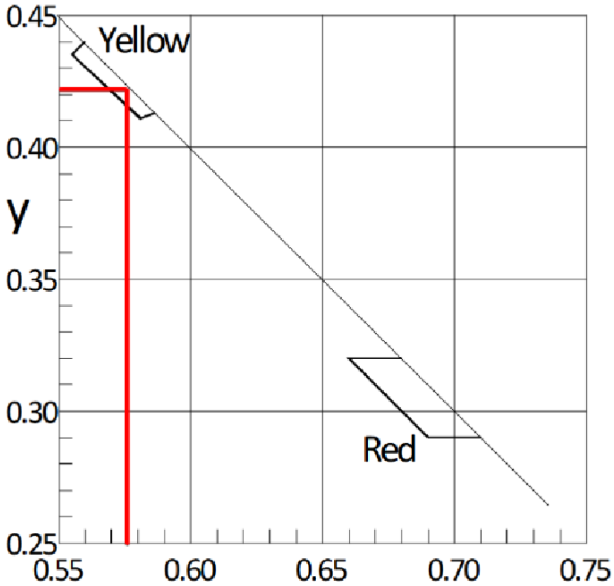

Figure 2.1: Calculated X and Y for 590 nm [**luxalight_cie_convertor**].

1. **Group-occulting:** This setting provides onlookers greater periods of light than darkness for easier identification. The IALA specification for special marks states "the number of eclipses in a group should not be greater than four in general, and should be five only as an exception"[12].
2. **SeaLabMOS group-occulting:** To better meet endurance needs and to help the LED not overheat, the flashing pattern is only shown once per minute (user configurable).

2.7.3. Automatic Timing of the Light

While "time switches, GNSS synchronization, [or] a real-time clock" *can* control light operation, the predominant method for night operation of AtoN lights is via photo-sensitive devices[13]. This allows automatic adaptability during poor visibility, such as fog. Care must be taken to select the correct ambient light switching level to minimize energy consumption and material wear. Ambient light levels (in lux) for different conditions are stated in the document's Table 1, and astronomical events with associated lux are listed in Table 2 [**iala_levels_g1038_2016**].

In the current state of the SeaLabMOS, there is no light sensor integrated. We are investigating two options that could suit this use-case.

1. Adafruit TSL2591 High Dynamic Range Digital Light Sensor.
2. Adafruit AS7262 6-Channel Visible Light / Color Sensor Breakout.

While the light sensor could be used to control the beacon on a surface asset, we are also investigating the potential for light sensors to simulate PAR sensors.

2.8. Light Sensors

While the light sensor could be used to turn the LED beacon on and off, we are also investigating the potential for use as a low-cost PAR sensor.

– Adafruit TSL2591 High Dynamic Range Digital Light Sensor (Product ID: 1980). Ultra low-light sensor with a wide dynamic range. I2C interface and STEMMA QT.
– Adafruit AS7262 6-Channel Visible Light / Color Sensor Breakout (Product ID: 3779). Provides six discrete wavelength channels (450–950 nm). I2C communication and STEMMA QT.

This is slightly outdated, but Chapter 4 is currently the focus. If curious, research is ongoing. If you have any thoughts, suggestions, ideas, or other thoughts, please reach out!

[12] **iala__characters_r0110_2021**.
[13] **iala_levels_g1038_2016**.

3

Sensors

3.1. Battery Voltage

Connect a 100KΩ resistor between BAT and A2 and a second 100KΩ from A2 to GND, following the power management instructions from Adafruit: https://learn.adafruit.com/adafruit-feather-rp2040-adalogger?view=all.

Table 3.1: Battery voltage divider connections

From	To
VBAT	Analog A2
Analog A2	Ground

Voltage is measured using a resistor voltage divider connected to the Feather's analog input pin as described by Adafruit[1]. The Feather's ADC measures a range of 0–3.3 V, so the voltage divider scales the battery's max of 4.2 V down to a range compatible with the ADC. The divider consists of two resistors, R_1 from the battery positive to the analog pin, and R_2 from the analog pin to ground. The voltage at the analog pin is given by:

$$V_{\text{ADC}} = V_{\text{bat}} \cdot \frac{R_2}{R_1 + R_2}$$

where V_{bat} is the battery voltage. The ADC reading is then converted to an actual voltage by scaling the raw value to the 0–3.3 V range of the ADC and then multiplying by the inverse of the voltage divider ratio. In the firmware, this is implemented as:

```
float v = analogRead(BATTV_PIN); // read ADC
v /= 4095.0;  // convert to 0-1
v *= 3.3;     // scale to voltage at analog pin
v *= 2.0;     // undo 1:1 voltage divider
```

3.1.1. Voltage Correction Factor

Each system built will have a different system-specific correction factor that must be applied to account for individual resistor tolerances. This factor is determined empirically by comparing the feather's voltage reading to a multimeter's measurement. In the firmware this correction implemented within *battMonitor.h* as:

Figure 3.1: Resistors for voltage measurement and small wire for RTC interrupt

[1]**adafruit_feather_rp2040**.

```
#if SYSTEM_NAME == GREEN (for example)
  v *= 1.053;
```

Steps to Find Correction Factor:

1. Ensure the Feather is printing the voltage to the serial monitor, for example with this program[2].
2. Measure from the positive wire of the battery (once you've wired up the power switch this should be easily accessible) to the Feather's ground
3. Update the relevant SYSTEM_NAME's correction factor until the Feather matches the reading

For example, if the multimeter reads $V_{\text{mm}} = 3.53$ V and the Feather reads $V_{\text{Feather}} = 3.67$ V, the correction factor can be calculated as

$$\text{Correction factor} = \frac{V_{\text{Feather}}}{V_{\text{mm}}} = \frac{3.67}{3.53} \approx 1.043.$$

3.2. Salinity: Atlas Scientific EZO-EC Embedded Conductivity Circuit and K1.0 Mini Probe

The SeaLabMOS family uses the Atlas Scientific EZO-EC conductivity circuit with the Mini Conductivity K 1.0 kit[3] to measure salinity via electrical conductivity. The EZO-EC circuit supports multiple output formats and is compatible with 3.3 V and 5 V systems, making it ideal for embedded applications. When paired with an accurate temperature sensor, this circuit allows real-time, in-situ salinity calculation using the Practical Salinity Scale (PSS-78).

3.2.1. Wiring

The EZO-EC circuit supports UART and I^2C, but the SeaLabMOS uses UART for direct communication with the Feather RP2040 Adalogger wired as follows:

Table 3.2: Sensor to carrier board wiring

Module Pin	Feather
VCC	3.3 V
GND	Ground
TX	RX (e.g., `Serial1 RX`)
RX	TX (e.g., `Serial1 TX`)
OFF	GPIO digital pin D4

Physically, this looks like using a 5-pin JST SM[4] plug to make the datalogger tray capable of quick disconnection from the sensor tray. First, solder one end of the plug to the carrier board. Then, the other into the feather. Ensure you have the right orientation of the plug, because it can only plug in one way.

[2] https://github.com/sscopazzi/SeaLabMOS/tree/main/single_sensor_programs/vBatt_only
[3] https://atlas-scientific.com/kits/mini-conductivity-k-1-0-kit/
[4] https://www.adafruit.com/product/1664

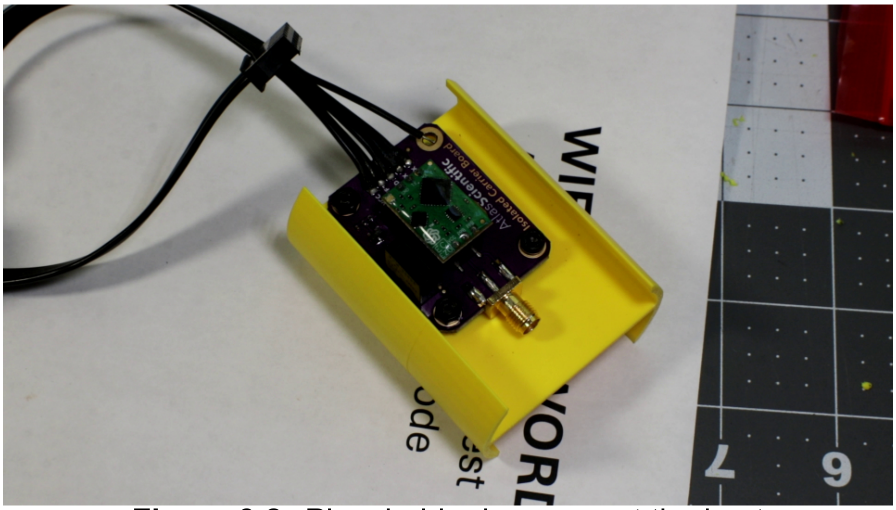

Figure 3.2: Placeholder image - not the best

For the feather side, start with one end of the plug's wires. They will end up twisting at interesting angles to get situated. This might be the trickiest part of the soldering required so take your time.

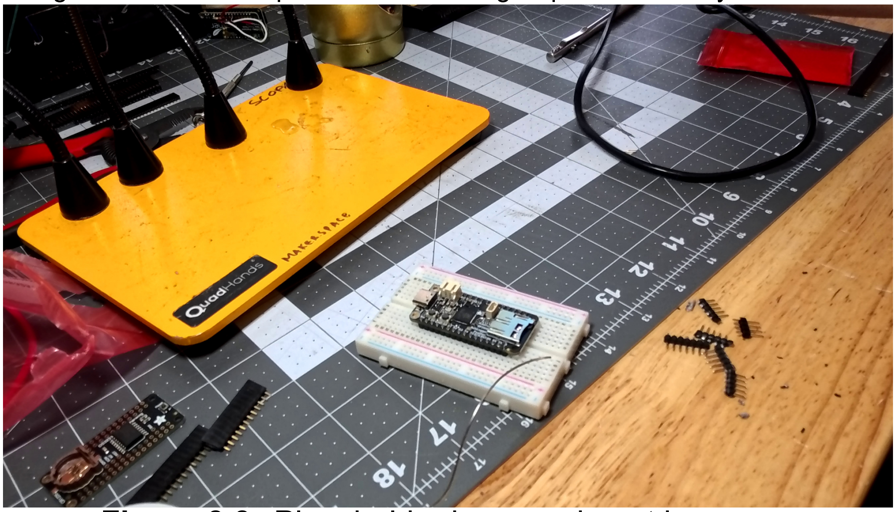

Figure 3.3: Placeholder image - do not have one

3.2.2. Calibration

By default on arrival, the EZO-EC should be configured for UART communication. If not, it can be configured for UART by sending the command `I2C,0` over a temporary I^{2}C connection, or by holding the EZO-EC's `PGM` pad low on power-up for >10 seconds, which resets it to UART mode.

Once wired, it is possible to "talk" with the EZO-EC circuit through the firmware and Arduino IDE. Once the Feather is wired to the sensor as above, it's possible to send the commands listed in the EZO-EC datasheet[5]. The EZO-EC circuit supports one-, two-, or three-point calibration. For highest accuracy across a range of salinity, the SeaLabMOS uses the three-point calibration: dry, low, and high. Before calibrating the probe, set the temperature compensation value to the temperature reported by the Blue Robotics Fast Temp using the command:

```
RT,<value>\r
```

The carriage return ("`\r`") is selected in the Arduino IDE and does not need to be manually typed. This will be used throughout the calibration process, which is as follows:

1. **Dry Calibration**:
   – Remove the probe from all liquids and ensure it is thoroughly dry.

[5] `https://files.atlas-scientific.com/EC_EZO_Datasheet.pdf`

- Once powered on, allow the probe to stabilize in air away from other electronics for 10–15 seconds.
- Send `Cal,dry` via UART.

2. **Low-Point Calibration**:
   - If wet and/or exposed or unknown previously, rinse the probe with distilled water and gently blot dry and/or blow dry with compressed air.
   - Type, but DO NOT send, the command that you will send (e.g., `Cal,low,12880`).
   - Immerse the probe in the low-point calibration solution (e.g., 12,880 µS/cm).
   - Wait for readings to stabilize (usually about 5-10 seconds). You'll notice that if left in the calibration solution for too long, the values will slowly drift. Send the calibration command once the value stabalizes, but before it drifts.
   - Send the command `Cal,low,<value>` (e.g., `Cal,low,12880`).
3. **High-Point Calibration**:
   - Rinse again with distilled water and blot the probe dry and/or blow dry with compressed air.
   - Type, but DO NOT send, the command that you will send (e.g., `Cal,low,80000`).
   - Place it in the high-point calibration solution (e.g., 80,000 µS/cm).
   - After readings stabilize, send `Cal,high,<value>` (e.g., `Cal,high,80000`).

Once complete, check the calibration status by sending `Cal,?`. A properly calibrated circuit will return the calibration points currently in use. If you followed the above, it will return 3.

3.2.3. Atlas-Scientific Power Management

To minimize power consumption during long deployments and when not actively sampling, the SeaLabMOS firmware turns the circuit on and off using GPIO Digital Pin 4.

3.3. Temperature and Pressure Sensor Wiring

Find the section that pertains to the sensors you have selected.

3.3.1. Blue Robotics Temp and Pressure

From Blue Robotics, the sensors has a 4-pos JST GH that is compatible with most flight controller boards, including Pixhawk and others (https://bluerobotics.com/store/sensors-cameras/sensors/bar30-sensor-r1/) (BM04B-GHS-TBT). However, this connector is useless for our purposes. It must be cut off and replaced with a STEMMA QT / Qwiic JST SH 4-pin Cable.

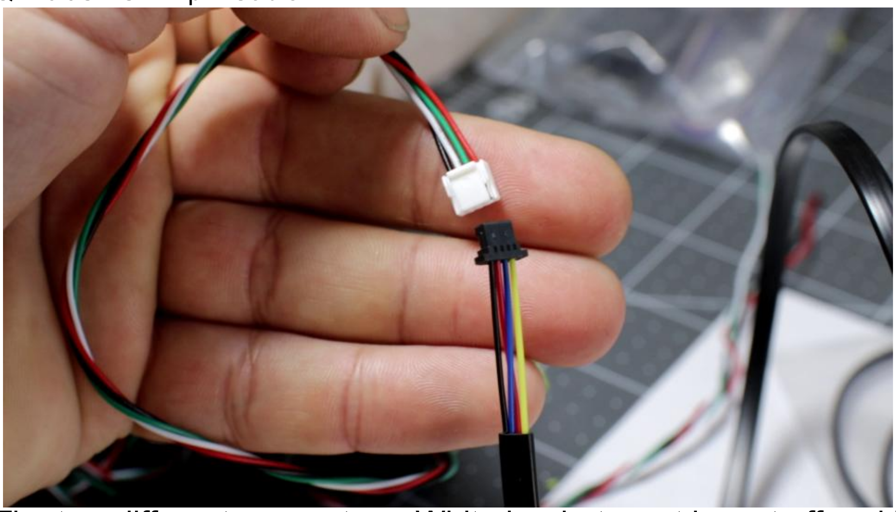

Figure 3.4: The two different connectors. White is what must be cut off and black is what it is exchanged for

Use the table below, or check the manufactures websites for which color is connected to what.

Table 3.3: Blue Robotics to Adafruit STEMMA QT cables

Pin / Wire	Blue Robotics	Adafruit STEMMA QT
1	Red – Vin	Red – V
2	Green – SCL	Yellow – SCL
3	White – SDA	Blue – SDA
4	Black – GND	Black – GND

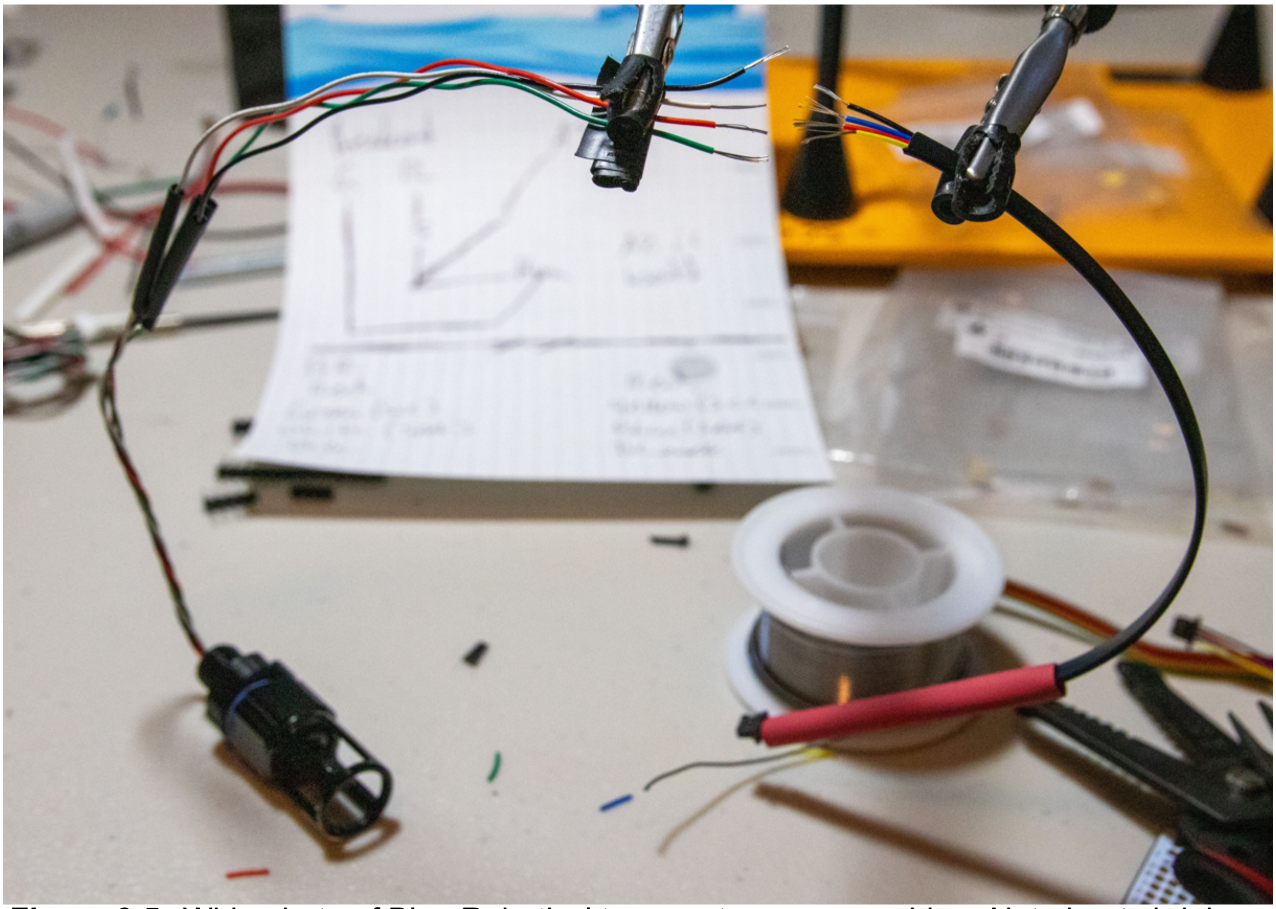

Figure 3.5: Wide photo of Blue Robotics' temperature sensor wiring. Note heat shrink on wires *before* soldering wires together

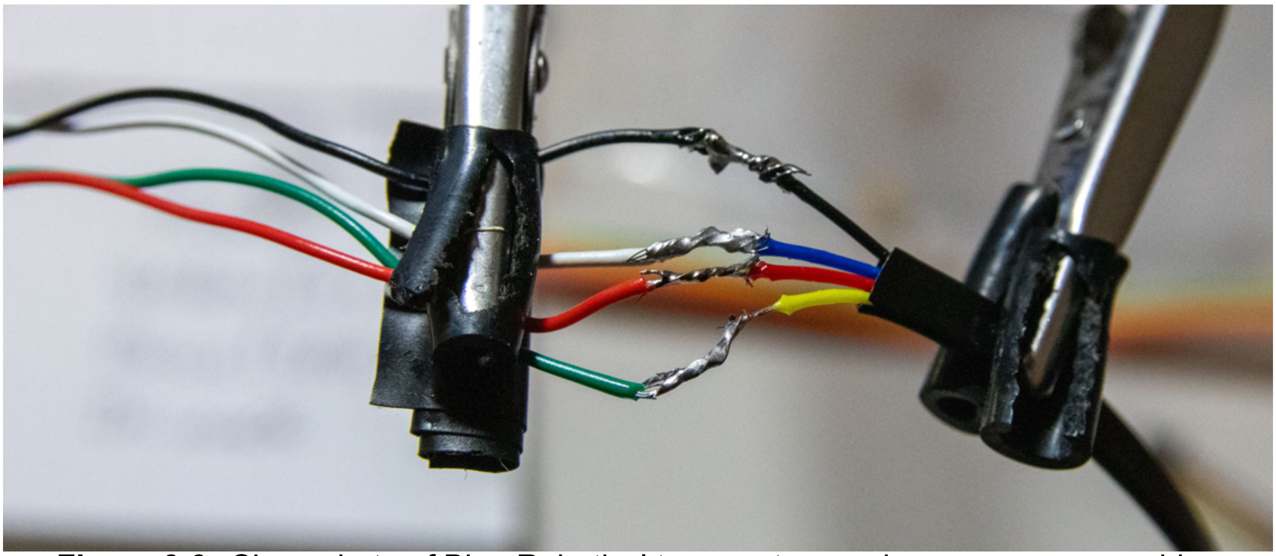

Figure 3.6: Close photo of Blue Robotics' temperature and pressure sensor wiring

3.4. Thermistor

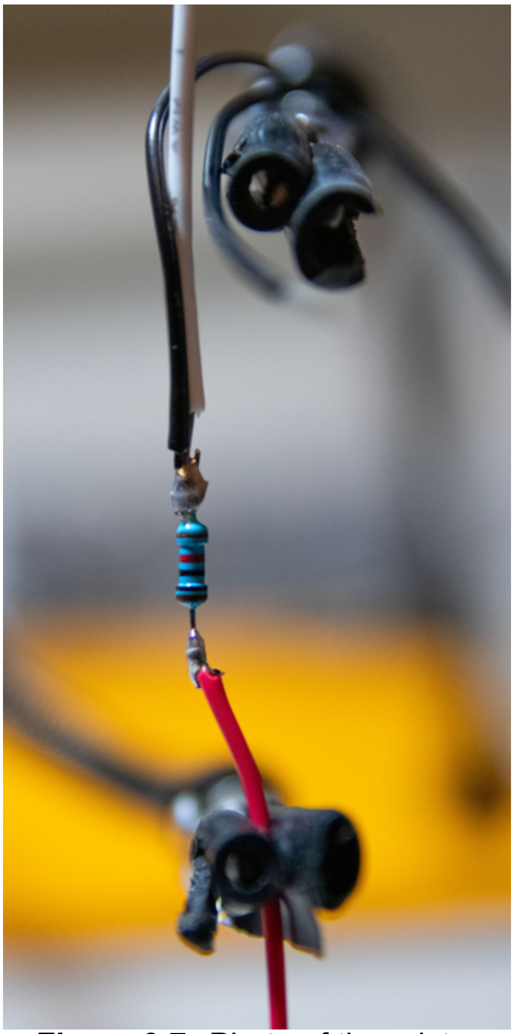

Figure 3.7: Photo of thermistor wiring. White wire goes to Feather's analog port

The thermistor has two wires and comes with a 10 kΩ resistor. The sensor wire (white in the photo) is put on the thermistor side of the resistor that is connected to 3.7V power. Wiring is as follows:

Table 3.4: Thermistor wiring. Wire 1 and Wire 2 from the thermistor are interchangeable

Feather	Thermistor
A3	Sensor wire (white in photo)
GND	Thermistor wire 1
3.3 V	Sensor wire and thermistor wire 2

3.4.1. Resistor Accuracy

The thermistor comes with a 10 kΩ resistor used in series; however, it is not perfectly 10 kΩ. To achieve the highest accuracy the resistor should be measured with a multimeter and the value in the firmware updated for the SYSTEM_NAME found within the firmware's therm.h.

3.4.2. Analog to Digital Resolution

By default, the Feathers analog to digital converter (ADC) resolution is 10-bit with 1024 discrete levels. The Feather board does, however, have the capability for 12-bit resolution with 4095 discrete levels for greater accuracy. Setting the ADC resolution to 12-bit is specified in the firmware's .ino file.

3.5. DS18B20

Please see the Adafruit resources. Will flesh out more here as time is available. https://www.adafruit.com/product/381 https://www.adafruit.com/product/5971 https://learn.adafruit.com/adafruit-chainable-ds18b20-extender-breakout/pinouts

3.6. Bench Test Before Final Assembly

At this point, you should have:

1. The voltage resistors soldered in place
2. The interrupt wire from the RTC to the feather soldered in place
3. The Blue Robotics pressure sensor with a QWIIC connector instead of the one it came with
4. The Blue Robotics Fast Temp sensor with a QWIIC connector instead of the one it came with (or the other temperature sensor(s) you selected, like the thermistor or DS18B20)
5. The Atlas Scientific probe and carrier board, if using the salinity sensor

(1), (2) , and (3)
The next step is bench testing the system, before gluing everything into place.

4

Default Builds

You will need the following equipment

1. Superglue
2. 9.5 mm and 12 mm drill
3. Small adjustable wrench
4. More things

4.1. 1.5in or 2.0in PVC

3D print all parts in the CAD files section of the GitHub page. Follow the filenames for how many of each to print.

4.1.1. End Cap

1. (optional) Sand the 3D printed protected tongs smooth, clean of dust, then spray paint your color of choice. Bright colors (yellow, pink) stand out more in the water (no photo yet)
2. Sand the outside surface of the PVC end cap flush; remove the uneven surfaces from the lettering. It is best to use a course grit (80) then a fine grit (200) sandpaper. A palm sander can be used for the initial course grit pass. It must be smooth to allow an even surface for the gaskets on the Blue Robotics sensors

Figure 4.1: Smooth on the left, new on the right

3. Superglue the inside washer into the PVC end cap. This is to give the Blue Robotics sensors an even plane to screw onto. It *should* snap into place, so do not fully push it in until you have added the glue. Place it smooth side (3D printer bed) side up.

Figure 4.2: The washer should snap into place so do not push it all the way in until you place the Superglue under it

4. Place the drill guide inside the PVC end cap and clamp in a vice, if drilling by hand, or use drill press with a vice. Clamping the end cap in a vice should squeeze the drill guide into place so it doesn't shift as you drill. Even if it moves, ignore it as it jumps around and maybe breaks. **Do not** let it get caught on your fingers/hands if it flies upward while you drill downward. So long as it helped you drill in the right place, it has served it's purpose. This is also why you print more than one
5. Drill the holes. Start with any of them. Two 9.5 mm and one 12 mm. If the Atlas Sci probe doesn't go in the hole, ream it out just a little bit with the drill by hand. Only a little bit, then test fit. Repeat as necessary.

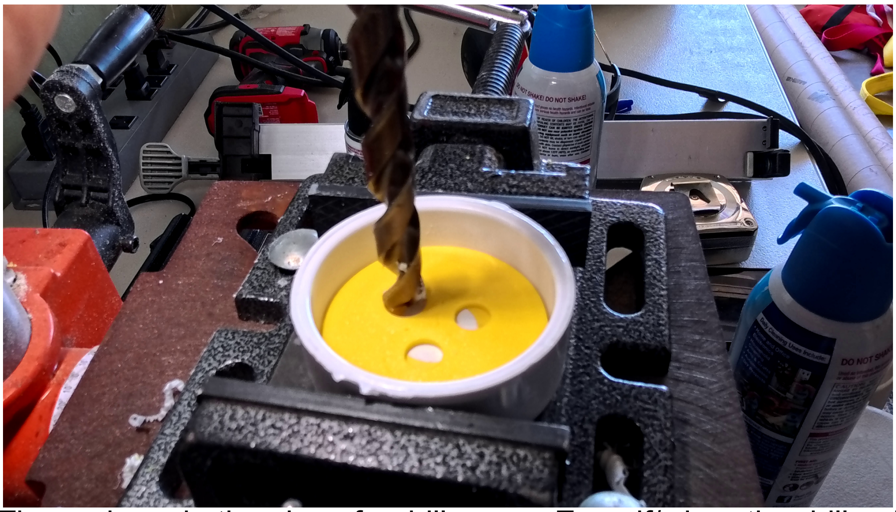

Figure 4.3: The end cap in the vice of a drill press. Even if/when the drill guide moves, it can break and the drill/end cap orientation does not change

6. The Blue Robotics temperature and pressure sensor recommend drilling a 10.2 mm hole for the 10 mm part. Instead, drill a 9.5 mm hole. Then, make sure the gasket is in the sensor, then screw it into the end cap until it is flush with the surface. Make sure to screw the Blue Robotics sensors in straight

Figure 4.4: Partly screwed into the PVC from the outside

Figure 4.5: Bolting into place from the back

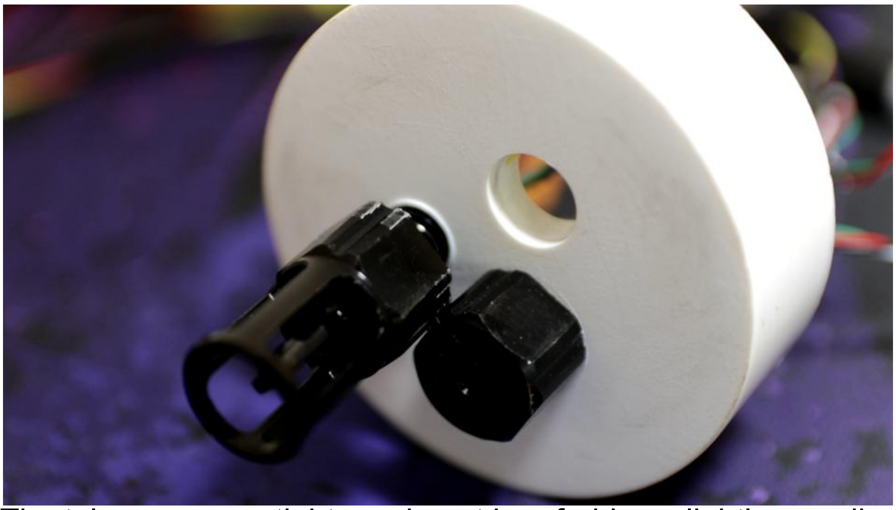

Figure 4.6: The tolerances are tight, so do not be afraid on slightly rounding the corners

7. Sand the part of the Atlas Sci probe that will get epoxied. Slide it into place where it will go and see how much to sand. Do not sand it only in one direction – all angles make a better rough surface finish

Figure 4.7: Smooth on the left, sanded on right

8. Situate the end cap with the now three installed sensors onto something that will keep it upright, sensor side down. A short section of 2 in PVC pipe works. Slide the epoxy ring into place to provide a dam to prevent the wet epoxy from falling off the inside raised circle into the slot where the PVC pipe will go. Resting is okay – no need to glue into place
9. Mix a 1/2 tablespoon of epoxy and stir well with a small stick/tongue depressor/similar. Carefully let gravity drop it off into the epoxy ring around all the sensor parts.
10. Gently raise and lower and spin the Atlas Sci probe to ensure epoxy gets between the probe and PVC. Use the stick to arrange the epoxy into an even surface and to make sure there are as few air bubbles as possible. If you poke the bubbles, it seems to work well
11. Place a small amount of epoxy on the end of the Blue Robotics sensor, to help provide a little strain relief where the wires come out. If they break off, it is better that they do it further from the sensor. Let dry at least six hours, or overnight

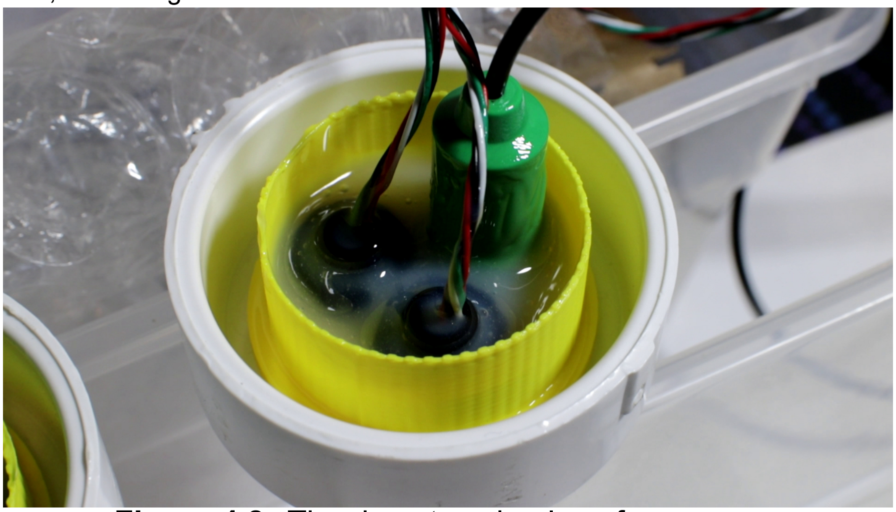

Figure 4.8: The ring stays in place forever now

12. Once dry, flip the end cap over and use Superglue to attach the protector tongs. Let that dry, then use epoxy like with the other side. Again, do your best to make no bubbles – poking them seems to work best. Fill in the area inside the protector tongs with epoxy. Let dry again for at least six hours

Figure 4.9: The top of the sensor face before epoxy. Glue the 3D printed tongs down with Superglue then put in epoxy like on the back

Figure 4.10: The end cap ready for bench testing after epoxy on both sides

4.1.2. System Tray

1. Glue the tray joiners to the system tray and tray extender. This is best done at the same time to ensure everything lines up. Place Superglue on the tray joiner and then hold both of them with the other two

parts. Kep holding it until the glue sets enough to let go

2. Glue the handle onto the flat part inside the tray extender
3. Glue the switch holder onto the handle, then put in the switch. Put the longer metal pokies on the back of the switch on the bottom side. The two short ones are for the LED in the switch, which we don't use here for better battery endurance

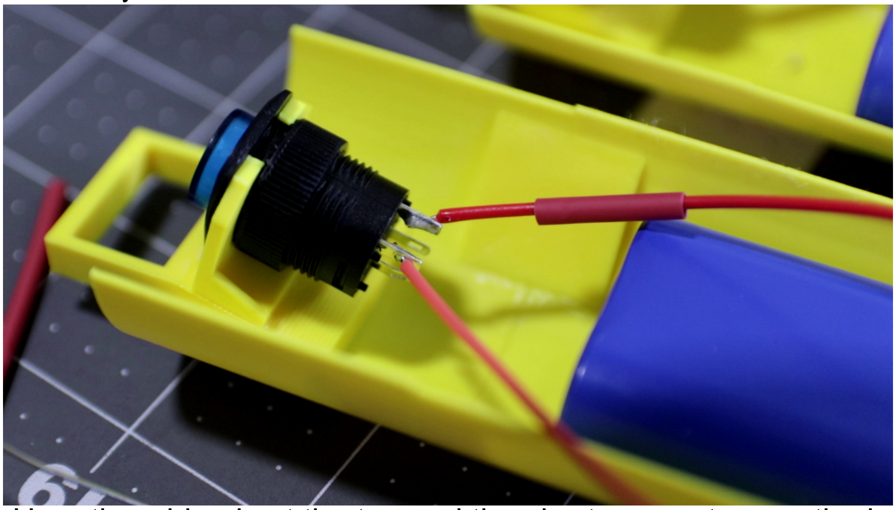

Figure 4.11: Here the wiring is at the top and the short connectors on the bottom are not cut; cut the short ones and solder to the long ones. The tray joiners are immediately to the left of the battery. Under the switch is the handle

4. Place the feather and RTC combo (with or without the OLED display) to get an idea of sizing for the battery. Either Superglue or Velcro tape the battery in place
5. Attach the battery's terminal to the feather, then cut the red wire. **Do not** cut the black one and **do not** but both simultaneously or you will short the battery
6. The batteries red wire is what gets switched. Solder a longer wire to the battery's red wire so it can reach the long metal pokie on the switch. Don't forget to use heat shrink on both ends
7. Solder a longer wire from the now cut red battery terminal plug in the feather to the switch. Again, don't forget the heat shrink

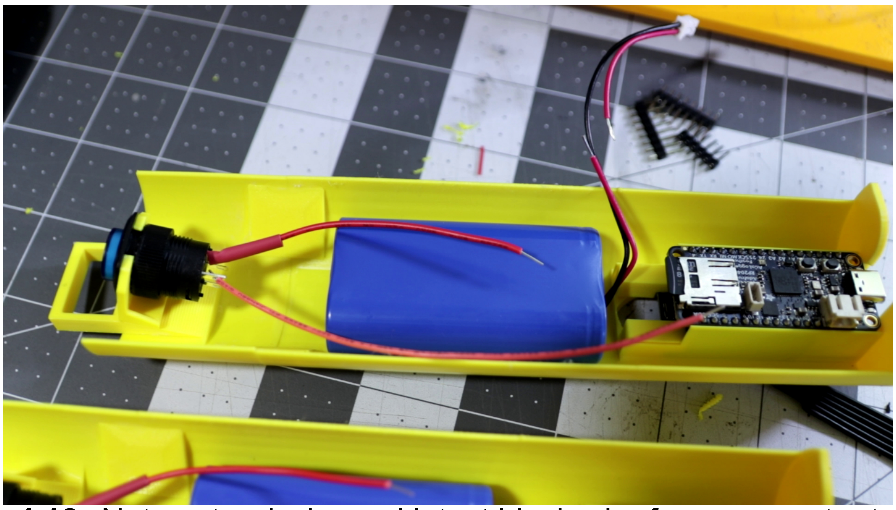

Figure 4.12: Note cut red wire and intact black wire from connector top right

4.1.3. Sensor Tray

The .stl file needs updating to add proper holes. This will get done by around November 2026 once Sophie is back ashore from sailing. In the meantime, drill holes in the proper locations.

1. Cut the pins off the bottom of the Atlas Scientific carrier board so the board will sit more flush on the sensor tray
2. Cut a piece or two of electrical tape to length and cover the bottom of the carrier board
3. Mark holes using the carrier board, then drill them
4. Mark holes for drilling for the Adafruit Qwiic / Stemma QT 5 Port Hub, then drill them
5. Use small bolts to attach into place

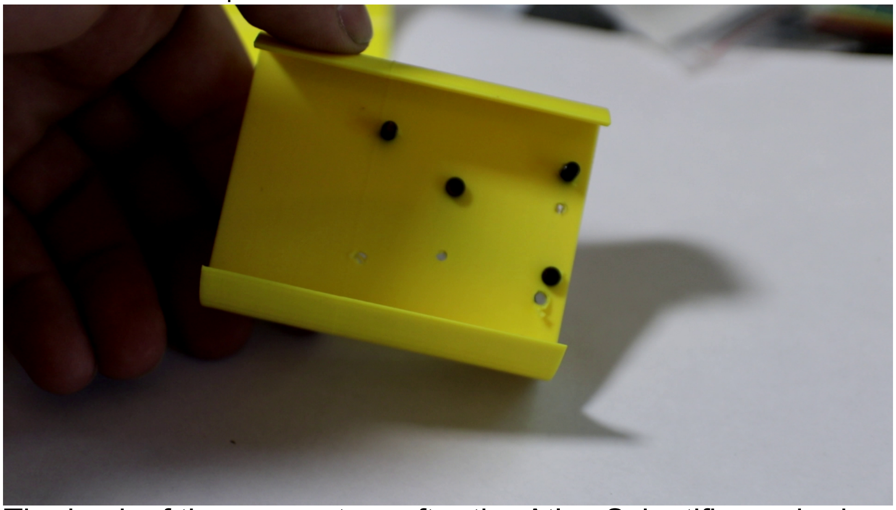

Figure 4.13: The back of the sensor tray after the Atlas Scientific carrier board was bolted into place, before the 5 port hub was added

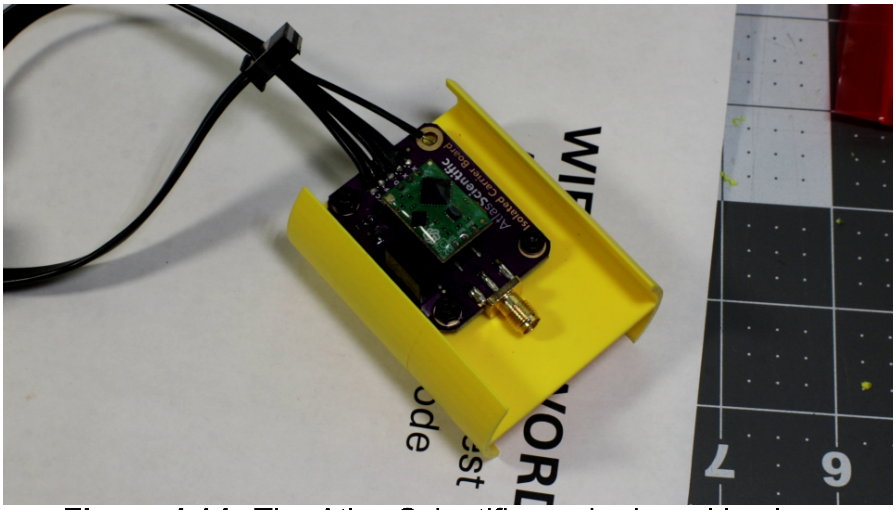

Figure 4.14: The Atlas Scientific carrier board in place

At this point, you should have an end cap with three sensors epoxied into place, a system datalogger tray, and the sensor tray. Before glueing the end cap into the PVC tube, do a bench test of the system.

Figure 4.15: Two CTDs ready for bench testing before glueing the end cap into the PVC pipe

4.2. Temp Logger

Cost How to assemble 3D printed parts and alternatives Endurance

4.3. Bottom Pressure Recorder

Cost How to assemble 3D printed parts and alternatives Endurance

4.4. Steel Pipe

Cost How to assemble 3D printed parts and alternatives Endurance

4.5. Surface Float (Nalgene)